

RAJA MOHAMMED

Senior Cloud & AI Automation Consultant | Enterprise Solution Architect

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EXECUTIVE SUMMARY

Enterprise technology consultant with **16+ years** architecting and delivering cloud, AI, and automation solutions for enterprise clients across Singapore and APAC region. Specialises in **production agentic AI** — most recently an Amazon Bedrock AgentCore platform automating the complete SOC/NOC incident response lifecycle across Microsoft and AWS — alongside multi-cloud architecture, Microsoft platform engineering, security automation, and enterprise data platforms.

Owns the **technical architecture and innovation roadmap** for multiple Cloud and AI initiatives within NEC's Innovation Team, providing technical direction to delivery teams of 4–10 solution architects, developers, infrastructure engineers, and consultants. Track record of measurable outcomes: analyst handling time reduced by an estimated **60–70%**, generative-AI inference cost cut **30–40%**, five SOC platforms consolidated into a single observability layer, and manual operational processes eliminated across security, IT, and business workflows.

CORE COMPETENCIES

Agentic AI & GenAI Architecture	Multi-Cloud Architecture (Azure · AWS)	Enterprise Solution Architecture
Security Automation & SIEM/SOAR	Cross-Cloud Enterprise Integration	Microsoft 365 & Power Platform
AI Observability & Incident Automation	Data Platform Engineering (Fabric)	Technical & Innovation Leadership

TECHNICAL SKILLS

AI & GenAI	Amazon Bedrock AgentCore (agent orchestration, conversation-context management, workflow coordination, automated actions); AWS Bedrock (Claude, Titan) — agent design, prompt engineering, model selection, token/cost optimisation; Retrieval-Augmented Generation (RAG), vector databases; enterprise AI observability; human-in-the-loop AI design; Microsoft Copilot Studio; Power Virtual Agents; n8n AI agent workflows
Cloud — AWS	Lambda, Bedrock, DynamoDB, Step Functions, API Gateway, SQS, S3, IAM, CloudWatch, Athena, Amazon Connect (CCP), GuardDuty; SigV4 authentication, Lambda proxy patterns, serverless architecture
Cloud — Azure	Logic Apps, Functions, Microsoft Sentinel, Entra ID/Azure AD, Azure AD B2C, Key Vault, Storage, Monitor/Log Analytics, Azure Arc (hybrid onboarding, Extended Security Updates)
Security & SIEM	Microsoft Sentinel (KQL detection engineering, analytics rules, playbooks, log onboarding); Securonix SIEM (Spotter API); Microsoft Defender; AWS GuardDuty; SolarWinds; MITRE ATT&CK-aligned detection logic; alert triage & false-positive reduction; Fleet MDM (osquery), endpoint vulnerability remediation
Identity & Access	Entra ID/Azure AD, Conditional Access, MFA, identity governance, Microsoft Graph delta sync, Azure AD SSO, ADFS, tenant-wide access remediation
Automation & Integration	Azure Logic Apps, Power Automate, n8n, RPA, Ansible; REST/JSON and Microsoft Graph APIs; webhook & event-driven integration; asynchronous webhook-callback architecture; cross-cloud Azure ↔ AWS integration
Microsoft 365 & Power Platform	SharePoint Online & On-Premises (2007–2019) administration, governance & migration (Sharegate, SPMT, AvePoint TTAM); Teams, Exchange Online, OneDrive, Intune, Windows 365/Cloud PC; Power Apps, Power Automate, Power Pages, Dataverse, Power BI (semantic models, RLS), Power BI Embedded; SPFx, CSOM/JSOM
Data & Analytics	Microsoft Fabric (Lakehouse, OneLake, Dataflow Gen2, PySpark, Spark SQL, Medallion architecture); data warehouse design, T-SQL stored procedures & reporting views; Power BI semantic modelling with Row-Level Security
Databases	Microsoft SQL Server/T-SQL, PostgreSQL, DynamoDB, Dataverse, Azure Storage Tables
Programming	Python, PowerShell (advanced), C#/.NET 8, JavaScript/TypeScript, Bash, Node.js, KQL

PROFESSIONAL EXPERIENCE

Senior Technical Consultant — Cloud & AI Automation

Sep 2021 – Present

NEC Asia Pacific Pte Ltd — Singapore (Remote) | Innovation Team

Owns the technical architecture and innovation roadmap for multiple Cloud and AI initiatives across the MIES managed-services platform serving APAC enterprise clients — spanning agentic AI, Microsoft Sentinel automation, AI-driven SOC/NOC platforms, cross-cloud Azure–AWS integration, contact-centre integration, and Microsoft Fabric data-platform delivery.

Agentic AI & GenAI Architecture

- Architected an **AI-driven SOC/NOC Incident Triage & Observability Platform** on **Amazon Bedrock** and **Amazon Bedrock AgentCore**, automating the complete incident response lifecycle across Microsoft and AWS environments — agent orchestration, persistent conversation context, workflow coordination, and automated action execution.
- Applied Amazon Bedrock LLM reasoning to security analysis, incident summarisation, risk assessment, and automated response generation within production SOC and NOC workflows.
- Designed AI-assisted user communication over Microsoft Graph and Microsoft 365 — response analysis, follow-up actions, SLA reminders, and automated incident closure or escalation.
- Implemented **enterprise AI observability**, enabling AI agents to monitor incident status, track workflow execution, and maintain end-to-end visibility across the incident lifecycle.
- Designed and deployed an AI-assisted triage workflow integrating **AWS Bedrock (Claude)** with **Microsoft Sentinel** for suspicious sign-in incidents, reducing analyst handling time by an **estimated 60–70%**.
- Applied model-selection and prompt-engineering techniques on AWS Bedrock to cut token consumption and inference cost by **30–40%** while preserving response quality.
- Built AI-powered conversational service desk and chatbot solutions on n8n with WhatsApp Cloud API and LLM backends — owning system-prompt design, conversation flow, and production deployment.

Cloud & Enterprise Integration Architecture

- Diagnosed an **AWS SigV4 authentication limitation** blocking Power Automate and Logic Apps from calling AWS services, and engineered a Lambda proxy integration pattern that unblocked AI agent calls from Microsoft workflows in production.
- Built an **asynchronous webhook-callback architecture** (AWS SQS + Step Functions) to bypass Power Automate's 2-minute action timeout, enabling reliable long-running AI and document-processing tasks.
- Architected **Amazon Connect (CCP) to Lambda integrations** enabling contact-centre agents to create and track incident tickets in-workspace across a **10-market APAC/ANZ/Greater-China rollout**.
- Designed a C# .NET 8 service implementing dual-delta sync against the Microsoft Graph /delta endpoint, synchronising Entra ID group membership into PostgreSQL with idempotent state handling at high volume.

Security Automation & Observability

- Engineered and tuned **Microsoft Sentinel KQL detections** for risky and new-geography sign-ins (MITRE T1078) with UPN and location filtering, improving SOC alert precision and reducing false positives.
- Unified **five SOC data sources** (Sentinel, Defender, AWS GuardDuty, Securonix, SolarWinds) into multi-tenant Grafana dashboards, delivering single-pane cross-cloud threat visibility across **12 documented initiatives in one fiscal year**.
- Built a Securonix SIEM custom playbook action in Python (Spotter API) handling token authentication, timezone normalisation, and policy-based violation filtering — automating incident enrichment and ticket creation.
- Architected a **NOC-to-SOC alert escalation pipeline** with documented swim-lane workflows, standardising alert handoff, classification, and case closure across operations and security teams.
- Built an automated firewall incident-response pipeline — SolarWinds detection to Grafana visualisation to webhook alert to automated incident creation to **Ansible-driven service restoration** — removing manual intervention from outage recovery.
- Architected Fleet MDM (osquery) automation orchestrated via Logic Apps with PowerShell remediation scripts, automating endpoint vulnerability remediation at scale across the managed-services estate.

Data Platform, Microsoft 365 & Governance

- Delivered an end-to-end **Microsoft Fabric data warehouse** (Discovery → Design → Build), centralising SharePoint Online, SQL Server, and Excel sources into OneLake via a Medallion architecture — 9 staging/transformation tables, 10 T-SQL stored procedures, reporting views, and a Power BI semantic model with RLS; authored the SOW, deployment work instructions, and handover documentation.

- Designed and shipped an **Employee Onboarding automation** on SharePoint, Power Apps, and Power Automate with a Microsoft Copilot Studio assistant — replacing a manual, email-driven process with governed auto-routing across four stakeholder groups.
- Authored advanced **PowerShell governance automation** to audit and remediate orphaned and resigned-user access across all SharePoint sites in the MSBU tenant, with structured compliance reporting.
- Integrated **Power BI Embedded** (Service Principal + Power BI JS SDK, custom Python backend) into the MIES customer-facing portal, giving external clients secure self-service reporting access.
- Led zero-data-loss SharePoint 2013 to SharePoint Online migrations using Sharegate and SPMT across multiple enterprise tenants, alongside Office 365 tenant setup and ADFS sync.

Technical Leadership

- Provide **technical direction to project teams of 4–10 members** — solution architects, developers, infrastructure engineers, and consultants — working alongside project managers and customer stakeholders.
- Influence enterprise architecture decisions, mentor engineers, and coordinate cross-functional collaboration across cloud, security, and operations teams.

Technical Consultant — Microsoft 365 & SharePoint

Feb 2017 – Aug 2021

NEC Asia Pacific Pte Ltd — Singapore (Onsite)

- **Led Microsoft 365 and SharePoint migration and modernisation workstreams across enterprise projects**, providing technical direction, solution architecture, delivery governance, mentoring, and client engagement.
- Delivered **zero-data-loss** on-premises to SharePoint Online migrations using Sharegate and SPMT across multiple enterprise tenants, with Office 365 tenant setup and ADFS synchronisation.
- Modernised legacy SharePoint JavaScript form solutions by migrating them to Power Apps, producing customer-facing migration risk and rationale documentation that **drove executive approval**.
- Contributed as a senior technical member of a multi-vendor delivery team on the **Fuji Xerox Document Management System transition spanning 8 countries** — recognised with NEC's Service Excellence Award.

Technical Consultant

Feb 2014 – Jan 2017

Maybank — Singapore

- Architected and deployed a **SharePoint 2013 farm** (Web Front-End, Database tier, Search, User Profile Services) for a top-tier ASEAN bank.
- Designed site collections, content types, document libraries, and **compliance-driven custom workflows** supporting audit and regulatory reporting requirements.
- Delivered custom web parts, master pages, and page layouts; led enhancements, defect resolution, and Agile delivery cycles.

Application Consultant

Nov 2012 – Jan 2014

NCS Pte Ltd — Singapore

- Delivered ASP.NET and SharePoint 2010 solutions for Singapore government and enterprise clients — web parts, features, application pages, and customised lists and libraries.
- Built identity-controlled portals and document management systems; led requirements gathering and stakeholder workshops.

Application Consultant

Mar 2012 – Nov 2012

Ogilvy & Mather — Singapore

- Architected and developed digital asset management modules in ASP.NET/SharePoint 2010 with NHibernate, later re-architected to Entity Framework.

Application Developer

Oct 2010 – Mar 2012

International SQLStar Pte Ltd — Singapore

- Built three-tier ASP.NET/MOSS 2007 applications with C# middleware, custom web parts, event handlers, and site definitions.

Application Developer

May 2009 – Jan 2010

RVMatrix Software Technologies — India

- Developed internet applications using HTML, JavaScript, IIS, COM, .NET, and XML; performed SQL Server query optimisation and system integration.

FEATURED PROJECTS

AI-Driven SOC/NOC Incident Triage & Observability Platform — Agentic AI

Enterprise agentic AI platform automating the complete incident response lifecycle across Microsoft and AWS. **Amazon Bedrock AgentCore** orchestrates AI agents, maintains conversation context, and drives automated actions; **Amazon Bedrock LLM** performs security analysis, incident summarisation, risk assessment, and response generation. Adds AI-assisted user communication with SLA reminders and automated closure or escalation, plus **enterprise AI observability** across the incident lifecycle.

Stack: Amazon Bedrock, Amazon Bedrock AgentCore, AWS Lambda, API Gateway, DynamoDB, SigV4, Microsoft Sentinel, Azure Logic Apps, Microsoft 365, Microsoft Graph

AI Security Operations Agent — Multi-Cloud GenAI Platform

Multi-Logic-App AI agent automating Microsoft Sentinel incident response — triage handler, analyst reply processor, and SLA monitor — with AWS Bedrock for decisioning, DynamoDB for state, and Microsoft Graph for analyst notification. **Reduced analyst handling time by an estimated 60–70%** and inference cost by 30–40%.

Stack: Azure Sentinel, Logic Apps, AWS Bedrock (Claude), Lambda, DynamoDB, SQS, Step Functions, Microsoft Graph

Multi-Country AI Contact Centre Rollout — Amazon Connect (10 Markets)

Contributed to an Amazon Connect-powered AI contact-centre rollout across **10 APAC, ANZ, and Greater-China markets**, integrating the MIES Service Portal, Teams chat app, and mobile self-service app. Built **CCP-to-Lambda integrations** enabling agents to create and track incident tickets in-workspace, eliminating context-switching.

Stack: Amazon Connect (CCP), AWS Lambda, Python, Office365-REST-Python-Client, SharePoint Online, Azure AD SSO

Microsoft Fabric Data Warehouse — Production Client Delivery

End-to-end engagement (Discovery → Design → Build) centralising SharePoint Online, SQL Server, and Excel sources into OneLake via a **Medallion architecture**: 9 staging and transformation tables, 10 T-SQL stored procedures, reporting views, and a Power BI semantic model with Row-Level Security. Authored the SOW and handover documentation.

Stack: Microsoft Fabric, OneLake, Dataflow Gen2, T-SQL, Power BI (semantic models, RLS), SharePoint Online

Cross-Cloud Integration Pattern — Microsoft to AWS

Diagnosed an **AWS SigV4 authentication limitation** preventing Power Automate and Logic Apps from calling AWS Lambda and API Gateway, then engineered a Lambda proxy pattern that resolved it — unblocking AI agent calls from Microsoft workflows in production.

Stack: Azure Logic Apps, Power Automate, AWS Lambda, API Gateway, SigV4, SQS, Step Functions

Multi-Source SOC Observability Platform (Grafana)

Unified **five security platforms** — Sentinel, Defender, AWS GuardDuty, Securonix, and SolarWinds — into multi-tenant Grafana dashboards, giving APAC enterprise stakeholders single-pane cross-cloud threat visibility; migrated from development to production.

Stack: Grafana, AWS Lambda, API Gateway, Azure Logic Apps, REST APIs

Enterprise Power Platform Governance & Ideation Program

Designed Power Apps canvas applications, **Power Pages portal authentication and authorisation architecture**, Power Automate governance flows, Azure AD B2C identity controls, and Dataverse data modelling for an enterprise innovation-ideation platform, within a multi-vendor delivery team.

Stack: Power Apps, Power Pages, Power Automate, Dataverse, Azure AD B2C, SharePoint Online

Employee Onboarding Automation — Copilot Studio & Power Platform

Replaced a manual, email-driven onboarding process with a governed workflow: a Power Apps intake form over SharePoint Online, Power Automate routing and approvals across **four stakeholder groups**, automated reminders, and a **Microsoft Copilot Studio assistant** handling onboarding FAQs.

Stack: Microsoft Copilot Studio, Power Apps, Power Automate, SharePoint Online, Microsoft 365

EDUCATION & CERTIFICATIONS

Master of Computer Applications (MCA) — Alagappa University, India

2006 – 2009

Bachelor of Computer Science — Alagappa University, India

2003 – 2006

- **Claude with Amazon Bedrock** — Anthropic / Skilljar Certification (2026)
- **Microsoft Certified: Azure Solutions Architect** (AZ-300)
- **Microsoft Certified Technology Specialist (MCTS)** — SharePoint 2010, Application Development
- **ISO 20000 Service Management System (SMS) Requirements** — Training Completed
- **Service Excellence Award** — NEC Asia Pacific Pte Ltd, Fuji Xerox Document Management System Transition Project spanning 8 countries